Technical report: Policy as code for digital sovereignty
Taniya Offergeld, L00205444

Findings

- Free HCP limits the assignment in terms of creating policies and wether they are hard/soft/advisory coded
- Designing individual policies can be contradictory between sovereignty and resilience. In my point of view the data classification is important to know where to scale up on sovereignty and where redundancy is more important. Data at rest and data in flight encryption are useful tools here to balance resilience and data sovereignty.
- A region policy is not sufficient on its own. It only checks checks where, not who. It cannot show legal control, backups or admin access.

Recommendations

- Paid plan: policies synced from Git and several mandatory policies (to verify).
- Add infrastructure security policies (IMDSv2, encrypted EBS, open SSH). This important to protect data in approved regions.
- Treat sovereignty as more than residency: keys, operators, backups and exit plans.

What I would do again

- Test on the CLI first with real HCP mocks, then confirm in HCP. Having both, CLI and HCP, is great to learn how the terraform code and the policy code are connected.
- Before starting the task I completed the official beginner tutorial from HashiCorp called HashiCups. This was a great start to get comfortable with the syntax and files.

What I would do differently

- I would avoid vendor lock with Amazon. Migrating to different providers does not seem simple https://www.reuters.com/business/retail-consumer/amazon-microsoft-cloud-computing-services-should-fall-under-eu-tech-rules-eu-2026-06-25/ This article shows recent European Commission development on the topic. Being cloud agnostic or a mix of hybrid cloud usage/on premises services for sensitive data is recommended.
- Start decision log at the very beginning as the decisions change or fully understand after each step completed. The assignment is just a showcase of how pass/test policies work but in production environment the policies are more complex. A decision log is inevitable
- Learn how to implement stacks in terraform in addition to workspaces to create consistency across regions.